

Blood component ratios in children with non-traumatic life-threatening bleeding

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Abstract

Background and Objectives: In paediatric trauma patients, there are limited prospective data regarding blood components and mortality, with some literature suggesting decreased mortality with high ratios of plasma and platelets to red blood cells (RBCs) in massive transfusions; however, most paediatric massive transfusions occur for non-traumatic aetiologies and few studies assess blood product ratios in these children. This study's objective was to evaluate whether high blood product ratios or low deficits conferred a survival benefit in children with non-traumatic life-threatening bleeding.

Materials and Methods: This is a secondary analysis of the five-year, multicentre, prospective, observational massive transfusion epidemiology and outcomes in children study of children with life-threatening bleeding from US, Canadian and Italian medical centres. Primary interventions were plasma:RBC and platelets:RBC (high ratio $\geq 1:2$ ml/kg) and plasma and platelet deficits. The primary outcome was mortality at 6 h, 24 h and 28 days. Multivariate logistic regression models were used to determine independent associations with mortality.

Results: A total of 222 children were included from 24 medical centres: 145 children (median [interquartile range] age 2.1 years [0.3–11.8]) with operative bleeding and 77 (8.0 years [1.2–14.7]) with medical bleeding. In adjusted analyses, neither blood product ratios nor deficits were associated with mortality at 6 h, 24 h or 28 days.

Conclusion: This paper addresses a lack of prospective data in children regarding optimal empiric massive transfusion strategies in non-traumatic massive haemorrhage and in finding no decrease in mortality with high plasma or platelet to RBC ratios or lower deficits supports an exploratory analysis for mortality.

Keywords

haemorrhage, massive, non-traumatic, paediatric, ratios, transfusion

Highlights

- In children requiring massive transfusions for non-traumatic indications, neither high ratios of plasma and platelets to red blood cells nor low deficits of platelets and plasma are independently associated with decreased mortality at 6 h, 24 h or 28 days.

- Optimal empiric transfusion strategies for children with life-threatening haemorrhages may differ depending on aetiology: medical, operative or traumatic.
- Massive transfusions in children with non-traumatic bleeds are rare but associated with high mortality; thus, more studies are needed to understand the best blood product approaches.

INTRODUCTION

Life-threatening bleeding in children is uncommon but associated with significant morbidity and mortality [1–3]. Although traumatic injury accounts for more deaths in the paediatric population than any other cause, an international survey of predominantly large, academic medical centres found that more than half of paediatric massive transfusion protocol (MTP) activations stem from non-traumatic bleeding [4]. The initial paper from the massive transfusion epidemiology and outcomes in children (MATIC) study supported this finding in demonstrating 46% trauma, 34% operative and 20% medical indications [1]. The 28-day and in-hospital mortality rates after massive transfusion range from 35% to 65% depending on the aetiology of the bleed [1, 2, 5–7].

Despite the staggering mortality rate, the literature on transfusion strategies for paediatric non-traumatic life-threatening bleeding is sparse [4, 8, 9]. Current management of children with life-threatening bleeding is modelled after damage control resuscitation (DCR), a well-established strategy for haemorrhaging adult trauma patients. Its central tenet is balanced haemostatic resuscitation, either with a blood component-based approach with high ratios of plasma and platelets to red blood cells (RBCs) or low titre group O whole blood with minimization of crystalloids [10, 11]. DCR targets the lethal triad of hypothermia, acidosis and coagulopathy known to herald death in trauma resuscitation [11, 12]. Contrastingly, operative and medical bleeds begin in a controlled setting and include a variety of aetiologies, including gastrointestinal extracorporeal membrane oxygenation (ECMO) complications and cardiothoracic surgeries [1, 4]. Therefore, a one size fits all approach extrapolating adult trauma blood product studies to non-traumatic bleeds in children may not be effective.

Even in adults, only a small subset of massive transfusion studies extends beyond the trauma bay, predominantly to gastrointestinal bleeds and cardiothoracic surgeries. These studies suggest a possible benefit in short- [13] and long-term [14, 15] survival with higher plasma:RBC or platelet:RBC. One study in paediatric burn excisions found that higher ratios correlated with decreased blood product administration after surgery but did not evaluate mortality [16]. The primary objective of this secondary analysis of the MATIC study was to evaluate the independent association of ratios and deficits of plasma and platelets to RBCs to mortality in children with non-traumatic life-threatening bleeding to address this gap in paediatric massive transfusion literature.

MATERIALS AND METHODS

We conducted a secondary analysis of a prospective, multicentre, observational study of children with life-threatening bleeding at 24 children's hospitals in the United States, Canada and Italy between

January 2014 and October 2018. Children aged 0 through 17 were eligible if they received greater than 40 ml/kg of total blood products over 6 h (literature suggests 40 ml/kg as an inflection point, after which mortality increases in children with traumatic injury) [17]. Data extracted included demographics, blood type, vital signs, Glasgow Coma Scale (GCS), laboratory values related to coagulation and end-organ function, blood products and haemostatic adjuncts, Pediatric Risk of Mortality score (PRISM III), complications (acute respiratory distress syndrome, sepsis and acute kidney injury [AKI]) and mortality at 6 h, 24 h and 28 days.

This paper analysed children with non-traumatic bleeding, divided into operative (44% cardiothoracic, 15% neurosurgical, 14% general surgery, 7% liver transplant, 3% including otolaryngology/plastic surgery, obstetrics/gynaecology and orthopaedics, 8% 'other' and 14% procedural [biopsies and catheterizations]) and medical (32% gastrointestinal, 18% sepsis, 12% oncologic and 36% 'other') bleeds. Data regarding specific aetiologies described as 'other' were not recorded [1]. 69.9% of operative patients and 82.0% of medical were transfused under MTP [1]. Patients on ECMO were categorized as operative or medical by proximate cause of haemorrhage. Details regarding this cohort have been previously published [1]. An analysis of blood product ratios and deficits in children with traumatic injury from this dataset is detailed in a separate publication [18].

Data before the haemorrhage were gathered in the 24 h before the start of the bleed, and data during the event spans initiation (MTP activation or administration of first blood product if not under MTP) to the conclusion (MTP deactivation or last product with no more in the subsequent 60 min). Data collected after the haemorrhage span the 12 h after event conclusion. Only data collected during the life-threatening bleeding were used to calculate ratios and deficits.

Children were excluded if MTP was activated, but no blood products were transfused, or if they had multiple life-threatening bleeds. Patients were excluded from plasma:RBC analyses if plasma was not transfused, and from platelet:RBC analyses if platelets were not transfused. Blood product ratios were calculated by dividing plasma or platelet (in millilitres per kilogram) by RBC (in millilitres per kilogram). A high ratio was defined as >1:2 plasma:RBC or platelet:RBC as this cut-off is commonly used in transfusion studies, including the pragmatic, randomized optimal platelet and plasma ratios study [10, 19]. Product deficits were calculated as the volume (in millilitres per kilogram) of RBCs minus the volume of plasma or platelets.

Research ethics

This study was conducted under the waiver of informed consent granted by each participating centre's institutional review board.

Statistical analysis

Data were described as counts, percentages, medians and interquartile ranges as appropriate. Tests of association included chi-square for categorical measures and Mann–Whitney *U* for continuous. Many of the pretreatment measures considered as potential confounders had substantial amounts of missing data; thus, we could not meet the assumption of missing at random and did not perform multiple imputation models. Instead, we considered measures with no more than 10% of values missing and regressed these on each mortality endpoint. Any measure associated with mortality ($p < 0.2$) was considered for the multivariate logistic regression models unless there was evidence of collinearity ($r > 0.6$) with potential confounders. Because of listwise deletion, measures were excluded from the final model if including them reduced the sample size by more than 10%. Some categorical measures were excluded if there were too few events to estimate an odds ratio. Statistical significance is defined as $p < 0.05$. Analyses were conducted with Stata Statistical Software, Version 16 (StataCorp, College Station, TX, USA).

Covariates considered for inclusion in the operative model included age, sex, PRISM III, heart disease, mechanical ventilation, cardiac arrest, vasoactive drugs, upper gastrointestinal bleeding, renal disease, haemofiltration, and administration of any component, adjunct or fluid. In the medical cohort, the same measures were considered in addition to white blood cell count, haemoglobin, haematocrit and highest blood urea nitrogen.

RESULTS

A total of 222 children met inclusion criteria, with 145 (65%) experiencing operative bleeding and 77 (35%) medical.

Operative bleeding

Among children with operative bleeding, 67 (46%) had high-plasma ratios and 35 (24%) high-platelet ratios (Table 1). For children with operative bleeding, there were no differences in demographics between high and low plasma:RBC, but the high ratio group had lower platelet counts (98 vs. $155 \times 10^9/L$, $p = 0.04$) before the event (Table 1). The high-platelet ratio group was younger in median years (0.3 vs. 3.2) and was more likely to receive cryoprecipitate (71.4% vs. 31.9%) or recombinant factor VIIa (31.4% vs. 21.7%).

While there were no differences in complications between plasma high and low ratio groups, in the unadjusted analysis of platelet:RBC, the high ratio group had a higher incidence of AKI ($p = 0.003$) (Table 2).

Medical bleeding

Within the medical cohort, 53 (69%) had a high-plasma ratio and 21 (27%) a high-platelet ratio. Children receiving high plasma:RBC ratios for medical bleeding had lower pre-event GCS (12 vs. 14) (Table 3), and children who received high platelet:RBC were younger (2.2 years vs. 10) (Table 3).

The high platelet:RBC group had a higher median plasma:RBC ratio (Table 4), but children in the high plasma:RBC group did not have significant differences in the volume of platelets or median platelet:RBC ratio (Table 4). There were no differences in transfusion-related complications between the ratio groups.

Multivariate analysis

In the multivariate analysis, after adjusting for the potential confounders shown in the table, high ratios of plasma or platelets to RBC and low

TABLE 1 Demographics and severity of injury measures before life-threatening haemorrhage in children with operative bleeding

	Plasma:RBC (n = 145)			Platelet:RBC (n = 145)		
	High ($\geq 1:2$), (n = 67)	Low ($< 1:2$), (n = 78)	p-value	High ($\geq 1:2$), (n = 35)	Low ($< 1:2$), (n = 110)	p-value
Age (year) ^a	2.7 (0–12)	2.0 (0–13)	0.9	0.3 (0–8)	3.2 (0–13)	0.02
Male, n (%)	35 (52.2)	39 (50.0)	0.8	14 (40.0)	60 (54.5)	0.1
White, n (%)	44 (78.6)	43 (63.2)	0.1	24 (77.4)	63 (67.7)	0.4
Hispanic, n (%)	12 (19.7)	8 (12.3)	0.3	3 (9.7)	17 (17.9)	0.3
Pediatric Risk of Mortality III ^a	11.0 (6–2)	11.0 (6–17)	0.4	15.0 (7–21)	11.0 (5–17)	0.2
Age-adjusted hypotension, n (%)	38 (58.5)	40 (55.6)	0.7	22 (64.7)	56 (54.4)	0.3
Lowest temperature (°C) ^a	36.0 (35–37)	36.2 (36–37)	0.1	36.3 (35–37)	36.1 (36–38)	1.0
Lowest Glasgow Coma Scale ^a	14.5 (6–15)	10.0 (3–15)	0.3	15.0 (6–15)	12.5 (3–15)	0.2
Base excess (mmol/L) ^a	−0.3 (−4 to 5)	0.4 (−4 to 6)	0.7	1.2 (−4 to 5)	−0.6 (−4 to 7)	0.9
Lowest platelets ($\times 10^9/L$) ^a	98 (73–222)	155 (93–245)	0.04	117 (62–226)	136 (89–238)	0.3
Lowest haemoglobin (g/dl) ^a	9.6 (7–11)	10.0 (8–12)	0.08	10.5 (9–12)	9.6 (8–11)	0.06
Highest international normalized ratio ^a	1.5 (1–2)	1.4 (1–2)	0.3	1.4 (1–2)	1.4 (1–2)	0.9
Extracorporeal membranous oxygenation, n (%)	14 (18.2)	11 (12.6)	0.32	6 (15.8)	19 (15.1)	0.92

Abbreviation: RBC, red blood cell.

^aMedian (interquartile range).

Note: Bold value indicates the statistically significant.

TABLE 2 Interventions during and outcomes after massive transfusion by high and low plasma and platelets to red blood cells (RBCs) for children with operative bleeding

	Plasma:RBC			Platelets:RBC		
	High ($\geq 1:2$) (n = 67)	Low ($< 1:2$) (n = 78)	p-value	High ($\geq 1:2$) (n = 35)	Low ($< 1:2$) (n = 110)	p-value
Blood products/adjuncts						
RBCs (ml/kg) ^a	47.1 (25–75)	56.6 (24–109)	0.3	47.1 (22–75)	58.3 (28–103)	0.2
Plasma (ml/kg) ^a	39.1 (22–73)	9.04 (0–25)	<0.0001	33.5 (18–49)	19.6 (4–48)	0.06
Platelets (ml/kg) ^a	17.0 (7–38)	10.4 (0–28)	0.053	41.9 (23–65)	8.3 (0–21)	<0.0001
Any cryoprecipitate, n (%)	35 (52.2)	33 (42.3)	0.2	25 (71.4)	43 (31.9)	0.0008
Any recombinant factor VIIa, n (%)	12 (7.9)	13 (16.7)	0.84	11 (31.4)	14 (12.7)	0.01
Any tranexamic acid, n (%)	5 (7.5)	4 (5.1)	0.6	3 (8.6)	6 (5.5)	0.5
Complications						
Acute respiratory distress syndrome, n (%)	17 (25.4)	19 (24.4)	0.9	12 (34.3)	24 (21.8)	0.14
Sepsis, n (%)	4 (6.0)	9 (11.5)	0.2	2 (5.7)	11 (10.0)	0.4
Acute kidney injury, n (%)	23 (34.3)	19 (24.4)	0.2	17 (48.6)	25 (22.7)	0.003
Mortality						
Died within 6 h, n (%)	3 (4.5)	4 (5.2)	0.9	1 (2.9)	6 (5.6)	0.5
Died of haemorrhage within 6 h, n (%)	3 (100)	4 (100)		1 (100)	6 (100)	
Died within 24 h, n (%)	6 (9.1)	9 (11.7)	0.6	3 (8.6)	12 (11.1)	0.7
Died of haemorrhage within 24 h, n (%)	6 (100)	8 (88.9)	0.4	3 (100)	11 (91.7)	0.6
Died within 28 days, n (%)	14 (21.2)	19 (24.7)	0.6	7 (20.0)	26 (24.1)	0.6
Died of haemorrhage within 28 days, n (%)	14 (21.20)	19 (24.7)	0.6	7 (20.0)	13 (50.0)	0.7

^aMedian (interquartile range).

Note: Bold value indicates the statistically significant.

TABLE 3 Demographics and severity of injury measures before life-threatening haemorrhage in children with medical bleeds

Characteristic	Plasma:RBC (n = 77)			Platelet:RBC (n = 77)		
	High ($\geq 1:2$) (n = 53)	Low ($< 1:2$) (n = 24)	p-value	High ($\geq 1:2$) (n = 21)	Low ($< 1:2$) (n = 56)	p-value
Age (year) ^a	8.2 (2–15)	7.6 (1–15)	0.7	2.2 (0–13)	10.5 (2–15)	0.03
Male, n (%)	26 (49.1)	8 (33.3)	0.2	11 (52.4)	23 (41.1)	0.4
White, n (%)	34 (73.9)	11 (55.0)	0.3	13 (68.4)	32 (68.1)	0.8
Hispanic, n (%)	3 (6.5)	2 (9.1)	0.7	2 (10.5)	3 (6.1)	0.5
Pediatric Risk of Mortality III ^a	19.0 (12–24)	17.5 (11–23)	0.8	21.0 (16–27)	18.0 (12–23)	0.2
Age-adjusted hypotension, n (%)	31 (62.0)	19 (79.2)	0.1	15 (71.4)	35 (66.0)	0.7
Lowest temperature (°C) ^a	36.2 (35–37)	36.2 (35–37)	0.9	36.4 (35–37)	36.2 (36–37)	0.8
Lowest Glasgow Coma Scale ^a	12.0 (3–14)	15.0 (14–15)	0.045	13.5 (8–15)	13.5 (3–15)	0.9
Base excess (mmol/L) ^a	1.2 (–5 to 11)	0.2 (–16 to 5)	0.098	1.4 (–6–0.5)	1.1 (–7–0.13)	0.8
Lowest platelets ($\times 10^9/L$) ^a	38.0 (19–152)	85.0 (26–134)	0.1	35.5 (19–92)	54.5 (20–159)	0.3
Lowest haemoglobin (g/dl) ^a	8.0 (6–9)	8.2 (6–10)	0.4	8.6 (6–11)	7.4 (6–9)	0.2
Highest international normalized ratio ^a	1.9 (2–4)	1.5 (1–2)	0.1	1.9 (1–4)	1.7 (1–3)	0.5
Extracorporeal membranous oxygenation, n (%)	8 (14.3)	2 (7.1)	0.34	3 (13.0)	7 (11.5)	0.84

Abbreviation: RBC, red blood cell.

^aMedian (interquartile range).

Note: Bold value indicates the statistically significant.

deficits were not associated with mortality (Table 5). For both plasma and platelet ratios, PRISM III and ECMO were associated with increased mortality at 28 days. Although none of the ratio associations reached

statistical significance, all odds ratios for mortality in patients with medical bleeds were greater than one. In the operative cohort, the opposite non-significant trend was present: odds ratios less than one.

TABLE 4 Interventions during and outcomes after massive transfusion by high and low plasma and platelet to red blood cell (RBC) ratios for children with medical bleeding

	Plasma:RBC			Platelets:RBC		
	High (≥1:2) (n = 53)	Low (<1:2) (n = 24)	p-value	High (≥1:2) (n = 21)	Low (<1:2) (n = 56)	p-value
Blood products/adjuncts						
RBCs (ml/kg) ^a	27.3 (19–52)	36.2 (20–100)	0.2	22.9 (15–40)	28.8 (20–68)	0.5
Plasma (ml/kg) ^a	25.2 (16–46)	6.3 (0–22)	0.0004	27.5 (16–39)	18.3 (8–51)	0.3
Platelets (ml/kg) ^a	10.7 (4–20)	11.4 (4–21)	0.6	19.8 (14–36)	7.3 (0–16)	0.0001
Any cryoprecipitate, n (%)	21 (39.6)	8 (33.3)	0.6	9 (42.9)	20 (35.7)	0.6
Any recombinant factor VIIa ^a	9 (17.3)	4 (16.7)	0.9	4 (19.0)	9 (16.4)	0.8
Any tranexamic acid, n (%)	5 (9.4)	3 (12.5)	0.7	1 (4.8)	7 (12.5)	0.3
Platelet:Red blood cells ^a	0.3 (0.1–0.7)	0.22 (0.2–0.4)	0.2	0.77 (0.7–1.0)	0.20 (0.0–0.3)	<0.0001
Complications						
Acute respiratory distress syndrome, n (%)	13 (24.5)	5 (20.8)	0.7	6 (28.6)	12 (21.4)	0.5
Sepsis, n (%)	9 (17.0)	6 (25.0)	0.4	4 (19.0)	11 (19.6)	1.0
Acute kidney injury, n (%)	15 (28.3)	10 (41.7)	0.2	9 (42.9)	16 (28.6)	0.2
Mortality						
Died within 6 h, n (%)	15 (28.3)	5 (20.8)	0.5	6 (28.6)	14 (25.0)	0.8
Died of haemorrhage within 6 h, n (%)	15 (100)	4 (80.0)	0.08	6 (100)	13 (92.9)	0.5
Died within 24 h, n (%)	19 (35.8)	7 (29.2)	0.6	10 (47.6)	16 (28.6)	0.2
Died of haemorrhage within 24 h, n (%)	18 (94.7)	5 (71.4)	0.1	9 (90.0)	14 (87.5)	0.8
Died within 28 days, n (%)	36 (67.9)	13 (54.2)	0.2	15 (71.4)	34 (60.7)	0.4
Died of haemorrhage within 28 days, n (%)	23 (63.9)	6 (46.2)	0.3	9 (60.0)	20 (58.8)	0.9

^aMedian (interquartile range).

TABLE 5 Multivariate logistic regression for mortality outcomes at 6 h, 24 h and 28 days in children with operative and medical bleeding

Covariates	6-h mortality				24-h mortality				28-day mortality			
	Odds ratio	Confidence interval	p-value		Odds ratio	Confidence interval	p-value		Odds ratio	Confidence interval	p-value	
Operative												
Plasma/RBC ≥ 1:2	0.87	0.2–4.0	0.9	Plasma/RBC ≥ 1:2	0.76	0.3–2.2	0.6	Plasma/RBC ≥ 1:2	0.82	0.4–1.8	0.6	
Platelet/RBC ≥ 1:2	0.50	0.06–4.3	0.5	Platelet/RBC ≥ 1:2	0.75	0.2–2.8	0.7	Platelet/RBC ≥ 1:2	0.79	0.3–2.0	0.6	
Plasma deficit	0.99	0.99–1.0	0.8	Plasma deficit	1.00	0.99–1.0	0.4	Plasma deficit	1.00	0.99–1.0	0.2	
Platelet deficit	0.99	0.99–1.01	0.7	Platelet deficit	1.00	0.99–1.0	0.4	Platelet deficit	1.00	1.0–1.0	0.06	
Medical												
Plasma/RBC ≥ 1:2	1.50	0.5–4.7	0.5	Plasma/RBC ≥ 1:2	1.36	0.5–3.9	0.6	Plasma/RBC ≥ 1:2	1.79	0.7–4.8	0.2	
Platelet/RBC ≥ 1:2	1.20	0.4–3.7	0.8	Platelet/RBC ≥ 1:2	2.27	0.8–6.4	0.1	Platelet/RBC ≥ 1:2	1.62	0.5–4.8	0.4	
Plasma deficit	1.00	0.99–1.0	0.9	Plasma deficit	1.00	0.98–1.0	0.5	Plasma deficit	1.00	1.0–1.0	0.4	
Platelet deficit	1.00	0.99–1.0	0.6	Platelet deficit	1.00	1.0–1.0	0.8	Platelet deficit	1.00	0.99–1.0	0.8	

Note: Covariates in operative plasma and platelet ratio models: PRISM III, cardiac arrest, extracorporeal membrane oxygenation and congenital heart disease (CHD). Covariates in operative plasma and platelet deficit models: PRISM III, cardiac arrest, haemofiltration, any blood component, vasoactive drug, hypotension and CHD. Covariates in plasma and platelet ratio models: age, PRISM III, cardiac arrest, highest blood urea nitrogen and gastrointestinal bleed. Covariates in plasma and platelet deficit models: age, sex, highest respiratory rate, haemoglobin and PRISM III. Abbreviation: RBC, red blood cell.

DISCUSSION

This secondary analysis of the MATIC cohort found no improvement in 6-h, 24-h or 28-day mortality among children with

operative or medical bleeding when administered high plasma: RBC or high platelet:RBC, nor were there associations between mortality and lower plasma or platelet deficits. This is the first multicentre, prospective data to evaluate the effect of transfusion

ratios and deficits in children with non-traumatic life-threatening bleeding.

In a separate MATIC analysis of children with traumatic injury, higher plasma:RBC was independently associated with reduced mortality at 6 h and greater plasma deficits with increased mortality at 6 and 24 h [18]. The authors posited that the increase in 24-h mortality with greater plasma deficits without equivalent findings with ratios was due to ratios' inability to account for the product volume. In this analysis of non-traumatic bleeding, however, we did not find any difference in mortality analysed by ratios or deficits.

In trauma-induced coagulopathy, hypoperfusion and endothelial dysfunction from tissue damage constitute prime drivers in the complex cascade causing fibrinolysis, platelet dysfunction and systemic anticoagulation [19]. In operative bleeding, however, the extent of endothelial damage could be less pervasive, considering the controlled nature of surgeries. In non-traumatic bleeding, hypoperfusion is more likely to be rapidly corrected as opposed to traumatic hypoperfusion that can persist unattenuated in the field before medical aid arrival. The observed lack of mortality benefit with higher relative plasma and platelet volumes in non-traumatic haemorrhage may relate to the amelioration of these mechanisms that herald traumatic coagulopathy.

Operative bleeding is also likely addressed via immediate surgical haemostasis, which can outweigh the impact of blood components. Furthermore, the plurality of procedures in this cohort was cardiothoracic; thus, children might have been on cardiopulmonary bypass, often monitored intra-operatively with modalities like activated clotting times, anti-Xa assays and viscoelastic-based testing to guide tailored product selection [20].

By considering the heterogeneity of pathophysiologies and procedures in non-traumatic bleeding, a goal-directed, individualized approach could target clotting derangements and limit unnecessary product transfusion. Studies suggest a superiority of viscoelastic monitoring (VEM) to conventional coagulation tests in describing multifactorial clot formation [21]. In children, goal-directed studies are mostly perioperative, predominantly cardiothoracic, and suggest goal-directed resuscitation with thromboelastography (TEG) or rotational thromboelastometry (ROTEM) may improve morbidity [22] and reduce blood product volume [20, 23]. While adult mortality data in non-traumatic VEM-guided transfusions are limited and inconclusive [3, 24, 25], to the best of the authors' knowledge, no paediatric studies have evaluated mortality. Medical patients who were more thrombocytopenic at baseline but failed to improve with increased platelet:RBC or decreased deficits could benefit from these tests of platelet function. In our study, TEG/ROTEM data were insufficient to include in the analysis.

Children with medical haemorrhage had the highest mortality and the fewest patients. Even among adults, evidence is sparse and primarily limited to gastrointestinal bleeding, for which there is no consensus on blood components [9, 22]. One study in adults with non-traumatic haemorrhage demonstrated higher platelet and plasma ratios in patients with greater haemostatic lab abnormalities but no mortality association [26]. Our study's small sample size also makes interpretation of the trend towards increased mortality with high

ratios in medical patients and decreased mortality in operative patients difficult to interpret. A single-centre study of adult non-traumatic massive transfusion found an increase in mortality in medical and non-cardiothoracic, non-vascular surgery patients with low ratios [27]. A 2018 meta-analysis of plasma ratios included only two non-trauma studies, one cardiac surgery and one vascular, and showed lower ratios associated with greater mortality at 24 h and 30 days [28]. These discrepancies demonstrate the need for larger studies in both medical and operative paediatric life-threatening bleeding.

Regarding the increased incidence of AKI in children with operative bleeding receiving high platelet:RBC, the literature is mostly in cardiothoracic surgeries. Thrombocytopenia and AKI are common after cardiopulmonary bypass, and the magnitude of platelet nadir correlates with AKI severity [29]. Haematologic complications also nearly triple the odds of developing AKI after cardiopulmonary bypass in children [30]. Therefore, the relationship between high platelet:RBC and AKI is likely not causal but rather likely reflects thrombocytopenia associated with AKI leading to increased platelet administration.

Strengths of this study include its prospective design and inclusion of multiple large academic centres in and outside of the United States. Importantly, this is the first study to evaluate blood product ratios for life-threatening medical bleeding in children and the first multicentre study in paediatric operative bleeding. Furthermore, this study reports products in millilitres per kilogram instead of units, as this is standard paediatric dosing. Finally, although data were collected before, during and after transfusion, the analysis evaluated only blood products delivered during the life-threatening haemorrhage.

Limitations of this study include small sample sizes, particularly among medical patients, making multivariate analyses less reliable, and precluding analysis of the data by age groups to account for the range of haemostatic development from infancy to adolescence. Furthermore, although the data were remarkably complete for the most pertinent variables (blood products and mortality), some lab values were missing; thus, the covariates could not be evaluated. Among the incomplete data, cause of death other than haemorrhage was missing; therefore, we are unable to comment on deaths due to central nervous system or multiorgan failure. Other potential covariates, such as the use of anticoagulation and measures of cardiothoracic surgical complexity, were not included in the data collection. Finally, due to the observational nature of the data, we cannot determine causality.

This study suggests that in children with life-threatening bleeding, empiric massive transfusion strategies associated with decreased mortality may differ for trauma, operative and medical patients. It is difficult to make broad recommendations for management of life-threatening haemorrhage, particularly in patients with medical bleeds; thus, treatment should be individualized to the child's evolving disease process, medications and comorbidities. Future studies could evaluate mortality with goal-directed therapies using viscoelastic assays and explore the effects of haemostatic adjuncts, including factor VIIa, fibrinogen concentrates and cryoprecipitate. The pathophysiology of traumatic haemorrhage differs from that in operative and medical settings—two broad categories of bleeding which in and of themselves

have significant differences—and considering the high mortality with paediatric massive transfusions, more research, including clinical trials are warranted to identify blood product relationships to maximize survival in a child acutely decompensating due to non-traumatic exsanguination.

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CONFLICT OF INTEREST

Dr Spinella is a consultant for Secure Transfusion Services, Hemanext and Haima; Dr Leonard receives royalty payments from UpToDate; Dr Josephson is a consultant for Immucor, Octapharma and Cellphire, and has an unrestricted grant from Medtronics. The remaining authors have no conflicts to declare.

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